**BIUST — Department of Computing and Informatics**

# Group Project Assignment

*Guesthouse Discovery Platform — an Airbnb-style Application*

**Instructor:** Ontiretse Bagwasi **Group size:** As advised in class **Total marks:** 100

## Overview

In this group project, you will design and build a two-part application inspired by Airbnb, focused on discovering guesthouses rather than booking them. The system consists of a web application used by administrators to manage guesthouse listings, and a mobile application used by customers to browse and interact with those listings. There is no booking or payment functionality in this project — the focus is on listing management, discovery, search, and rating.

You are free to choose your own technology stack within the constraints stated below. Where a choice is left open, you should be able to justify it in your documentation and demo.

## Instructions

- For the web application, use React and TypeScript as the main language.
- For the backend, use any framework of your choice — Node, Laravel, etc.
- For the database, use any NoSQL database — Firebase, MongoDB, etc.
- Use GitHub for your version control.
- Feel free to be as innovative as possible.

## Part 1 — Web Application (Admin Portal)

### Technology requirements

- Frontend: React
- Database: a NoSQL database of your choice (e.g. MongoDB, Firebase Firestore, CouchDB)
- Backend: a backend language/framework of your choice (e.g. Node.js/Express, Django, Spring Boot, Laravel)

### Admin authentication and user management

- Admins must be able to log in to the web application
- A logged-in admin must be able to create additional users who also have login rights (i.e. admin accounts are not limited to a single hardcoded user)

### Guesthouse registration

Admins must be able to register a guesthouse, capturing at least the following information for each one:

- Name
- Logo
- Gallery (multiple images)
- Description
- Amenities
- Location as Google Maps coordinates
- Price per night
- City
- Country
- Contact details

You may add further fields beyond this list if you think they add value (e.g. number of rooms, check-in time, house rules), as long as the fields above are all present.

### Admin dashboard

The web application must include a dashboard that gives admins an overview of the platform at a glance, showing at least:

- The total number of guesthouses registered on the platform
- Which location (city) has the most registered guesthouses
- The most highly rated guesthouse

You are encouraged to add further statistics to the dashboard where you think they would be useful to an admin (e.g. average price per city, guesthouses added this month).

## Part 2 — Mobile Application (Customer App)

### Technology requirements

- Any language or framework of your choice (e.g. Flutter, React Native, Kotlin, Swift)
- The mobile app must read the same guesthouse data that admins register through the web application

### Browsing without an account

- Customers do not need to be logged in to view guesthouses
- Customers can search for guesthouses by location or by name
- Customers can sort search results using different filters, including price and rating
- When a search is performed, results should be sorted with the cheapest guesthouse displayed first by default

### Logged-in customer features

- Customers may create an account and log in
- Once logged in, a customer can save guesthouses as favourites
- Once logged in, a customer can rate each guesthouse
- There is no booking functionality anywhere in the app — customers can only view, search, favourite, and rate guesthouses

## Deliverables

- Full source code for the web application (frontend and backend), including your NoSQL database schema or collection structure
- Full source code for the mobile application
- A short README for each part explaining your tech stack choices, how to set up and run the project, and any environment variables or credentials needed (use placeholders, not real secrets)
- A brief data model / entity diagram showing how guesthouses, admins, customers, and ratings relate to one another
- A short live demo or demo video covering both the web admin portal and the mobile app

## Assessment Rubric

The detailed assessment rubric will be shared with you separately closer to the submission date.

## Group Work Guidelines

- Every member of the group is expected to contribute to both the web application and the mobile application, or to have a clearly documented division of labour
- Each group must submit a short contribution log stating which member worked on which parts of the system
- Group members may be asked individual questions about any part of the codebase during the demo, including parts a teammate wrote

## Submission Instructions

- Submit a link to your group's source code repository (e.g. GitHub), with both the web and mobile projects included
- Ensure your README instructions are complete enough for the instructor to set up and run both applications independently
- Only one submission per group is required, but every member's name and student number must be listed in the README